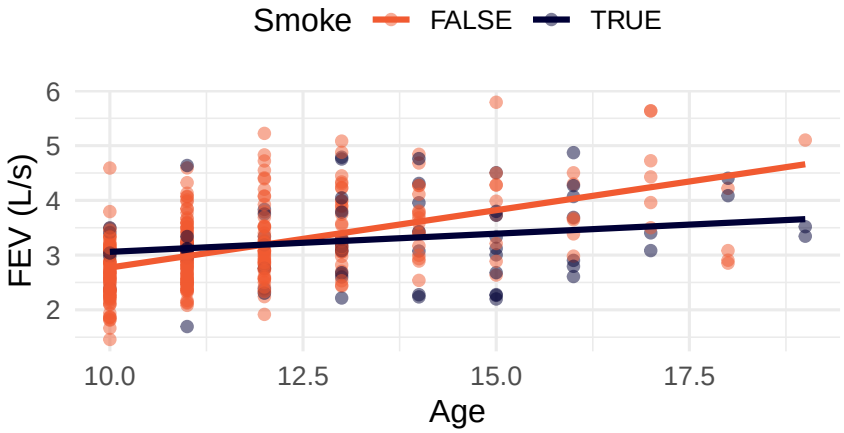
Day 12: Inference for linear combinations

We're interested in modeling **lung capacity** (FEV) on **age** and **smoking status** for a sample of adolescents. As kids get older, FEV generally increases, but we also think FEV decreases for smokers. Since smokers are likely to be older, we need a model that accounts for both.



Theoretical mean model:

$$\mu\{FEV \mid Age, Smoke\} = \beta_0 + \beta_1\, Age + \beta_2\, Smoke + \beta_3\, Age \times Smoke$$

term	estimate	std. error	t	p-value
Intercept	0.675	0.251	2.68	0.008
Age	0.210	0.021	9.97	<0.001
SmokeTRUE	1.720	0.563	3.05	0.002
Age:SmokeTRUE	-0.143	0.042	-3.40	<0.001

Example: Models for each group

Nonsmokers ($Smoke = 0$): $\mu\{FEV \mid Age\} =$

$\hat{y} =$

Smokers ($Smoke = 1$): $\mu\{FEV \mid Age\} =$

$\hat{y} =$

Reading the table: what is significantly different than 0?

- The intercept for non-smokers is _____
- The intercept offset for smokers is _____
- The slope for non-smokers is _____
- The *difference* in slopes between smokers and non-smokers is _____
- The slope for smokers is _____

The slope for smokers is $\beta_1 + \beta_3$. The table gives us $SE(\hat{\beta}_1)$ and $SE(\hat{\beta}_3)$, but **not** the standard error of their sum.

Why can't we just add $SE(\hat{\beta}_1) + SE(\hat{\beta}_3)$?

Why can't we use the p-value for `Age:SmokeTRUE` to test whether the smoker slope is zero?

> **Definition**
>
> A **linear combination** of coefficients has the form
>
> $$\gamma = c_1\beta_i + c_2\beta_j$$
>
> where c_1 and c_2 are constants
> **Estimate:** $\hat{\gamma} =$ _____
> **Standard error:**
>
> $$SE(\hat{\gamma}) = \sqrt{c_1^2\{SE(\hat{\beta}_i)\}^2 + c_2^2\{SE(\hat{\beta}_j)\}^2 + 2c_1c_2\,Cov(\hat{\beta}_i, \hat{\beta}_j)}$$

The extra term is there because $\hat{\beta}_i$ and $\hat{\beta}_j$ come from the *same* fitted model, so their estimation errors are not independent. $Cov(\hat{\beta}_i, \hat{\beta}_j)$ measures how they move together.

The covariance matrix of the coefficients

```
vcov(fev_lm) |> round(4)
```

```
                (Intercept)     Age SmokeTRUE Age:SmokeTRUE
(Intercept)          0.0632 -0.0052   -0.0632         0.0052
Age                 -0.0052  0.0004    0.0052        -0.0004
SmokeTRUE           -0.0632  0.0052    0.3172        -0.0234
Age:SmokeTRUE        0.0052 -0.0004   -0.0234         0.0018
```

The **diagonal** entries are the variances, $V(\hat{\beta}_j) = \{SE(\hat{\beta}_j)\}^2$.

- Check: $SE(\hat{\beta}_1) =$ _____ , so $\{SE(\hat{\beta}_1)\}^2 =$ _____

The **off-diagonal** entries are the covariances, $Cov(\hat{\beta}_i, \hat{\beta}_j)$. - The matrix is symmetric: $Cov(\hat{\beta}_i, \hat{\beta}_j) = Cov(\hat{\beta}_j, \hat{\beta}_i)$.

Example 1: CI for the smoker slope

Target: $\gamma = \beta_1 + \beta_3$. So $c_1 =$_____ and $c_2 =$_____

Estimate:

Standard error: pull the pieces out of the covariance matrix.

piece	value
$\{SE(\hat{\beta}_1)\}^2$	
$\{SE(\hat{\beta}_3)\}^2$	
$Cov(\hat{\beta}_1, \hat{\beta}_3)$	

90% CI: $\hat{\gamma} \pm t^*_{n-(p+1)} \cdot SE(\hat{\gamma})$, where $n = 345, p = 3$, so $df = 341$ and $t^* \approx 1.649$.

Interpretation:

> **What if we ignored the covariance?**
>
> Using only $\sqrt{\{SE(\hat{\beta}_1)\}^2 + \{SE(\hat{\beta}_3)\}^2}$ would give 0.047. The correct SE is 0.037. Here the covariance is **negative**, so ignoring it makes the interval too wide.

Testing a linear combination

Hypotheses: $H_0 : \gamma = \gamma_0$ vs. $H_a : \gamma \neq, <, \text{or} > \gamma_0$

Test statistic: $t = \dfrac{\hat{\gamma} - \gamma_0}{SE(\hat{\gamma})}$

Reference distribution: t distribution with $df = n - (p + 1)$

p-value: area in the tail(s) specified by H_a

Same four steps as a test for a single coefficient. Only the estimate and the SE change.

Example: Is the smoker slope different from zero? Using $\hat{\gamma} = 0.0664$ and $SE(\hat{\gamma}) = 0.0366$:

Your turn: smokers vs. nonsmokers at age 15

At age 15, how different is the mean FEV of smokers and nonsmokers?

0. Look back and the graph and identify what this question is asking about. What do you think you will find?

1. Write the theoretical mean function for each group evaluated at $Age = 15$, and subtract (smoker $-$ nonsmoker) to find γ.

2. The difference is $\gamma =$_____, so $c_2 =$ _____ $c_3 =$ _____

3. Find $\hat{\gamma}$

3. Write out $SE(\hat{\gamma})$ using the values from the covariance matrix.

4. Construct a 95% CI for γ ($df = 341, t^* \approx 1.967$).

5. Test $H_0 : \gamma = 0$ vs. $H_a : \gamma \neq 0$.

6. Interpret your results in context.

A shortcut: re-center the model

If we measure age as "years since age 15", the coefficient on Smoke *is* the smoker-minus-nonsmoker difference at age 15, and the standard table gives us its SE directly:

```
fev <- mutate(fev, Age15 = Age - 15)
lm(FEV ~ Age15 * Smoke, data = fev) |> tidy()
```

```
# A tibble: 4 x 5
  term            estimate std.error statistic   p.value
  <chr>              <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)        3.82     0.0780     49.0  2.33e-156
2 Age15              0.210    0.0210      9.97 1.01e- 20
3 SmokeTRUE         -0.430    0.125      -3.44 6.57e-  4
4 Age15:SmokeTRUE   -0.143    0.0422     -3.40 7.63e-  4
```

Compare the SmokeTRUE row to the output from the previous section.

Summary

To do inference for…	Use
a single coefficient β_j	t-test / CI from the coefficient table
a linear combination $c_1\beta_i + c_2\beta_j$	t-test / CI with $SE = \sqrt{c_1^2\{SE(\hat{\beta}_i)\}^2 + c_2^2\{SE(\hat{\beta}_j)\}^2 + 2c_1c_2\,Cov(\hat{\beta}_i, \hat{\beta}_j)}$
several coefficients at once ($\beta_2 = \beta_3 = 0$)	extra sums of squares F-test

- Never add standard errors. Use the covariance matrix.
- The reference distribution is always $t_{n-(p+1)}$ for a single linear combination.
- A linear combination is *one* number, so we use a t procedure. When $d > 1$ coefficients are tested at the same time, that's an F-test.

Your turn: in R

The R activity is posted on moodle. Here are some functions that might be helpful:

Function	Purpose
`coef()`	Extract the estimated coefficients from a fitted model (as a named vector)
`vcov()`	Extract the covariance matrix of the estimated coefficients from a fitted model
`x[i]`, `m[i, j]`	Pull off element `i` of a vector, or the entry in row `i` and column `j` of a matrix
`sqrt()`	Take the square root (e.g., to turn a variance into a standard error)